RF-05 and 06 where modified. RF-07 through 18 and NFR-06 through 07 are new

- RF-01: The system must allow a student to reserve an available sports court for an individual or informal activity.
- RF-02: The system must allow an official team captain to reserve a sports court.
- RF-03: The system must give official team captains priority when reserving courts before 6:00 PM.

- RF-04: The system must allow users to cancel their own reservation.
- RF-05: If a regular student cancels a reservation less than 2 hours before its start time, the system must record the action as a no-show.
- RF-06: If an official team captain cancels a reservation less than 24 hours before its start time, the system must record the action as a no-show

- RF-07: The system must allow Sports Complex administrators to create, remove, and manage available courts.
- RF-08: The system must allow Sports Complex administrators to resolve conflicts between reservations.
- RF-09: If a Sports Complex administrator forcefully cancels a reservation, the system must not apply any consequence to the user who created the reservation.

- RF-10: The system must allow official team captains to create recurring reservations for the same court, day, and time for at least 2 weeks and up to 8 weeks at a time.
- RF-11: A recurring reservation may include requests for specific equipment, such as balls or nets, and additional notes for the concierge.
- RF-12: If any occurrence of a recurring reservation is unavailable, the system must reject the entire recurring reservation.

- RF-13: The system must allow Sports Complex administrators to create reusable blocking templates containing a time, motive, and optional equipment.
- RF-14: The system must allow an administrator to apply a blocking template to any court and time.
- RF-15: Applying a block must override any preexisting reservation for the selected court and time.
- RF-16: If a block overrides an existing reservation, the system must notify the reservation creator.

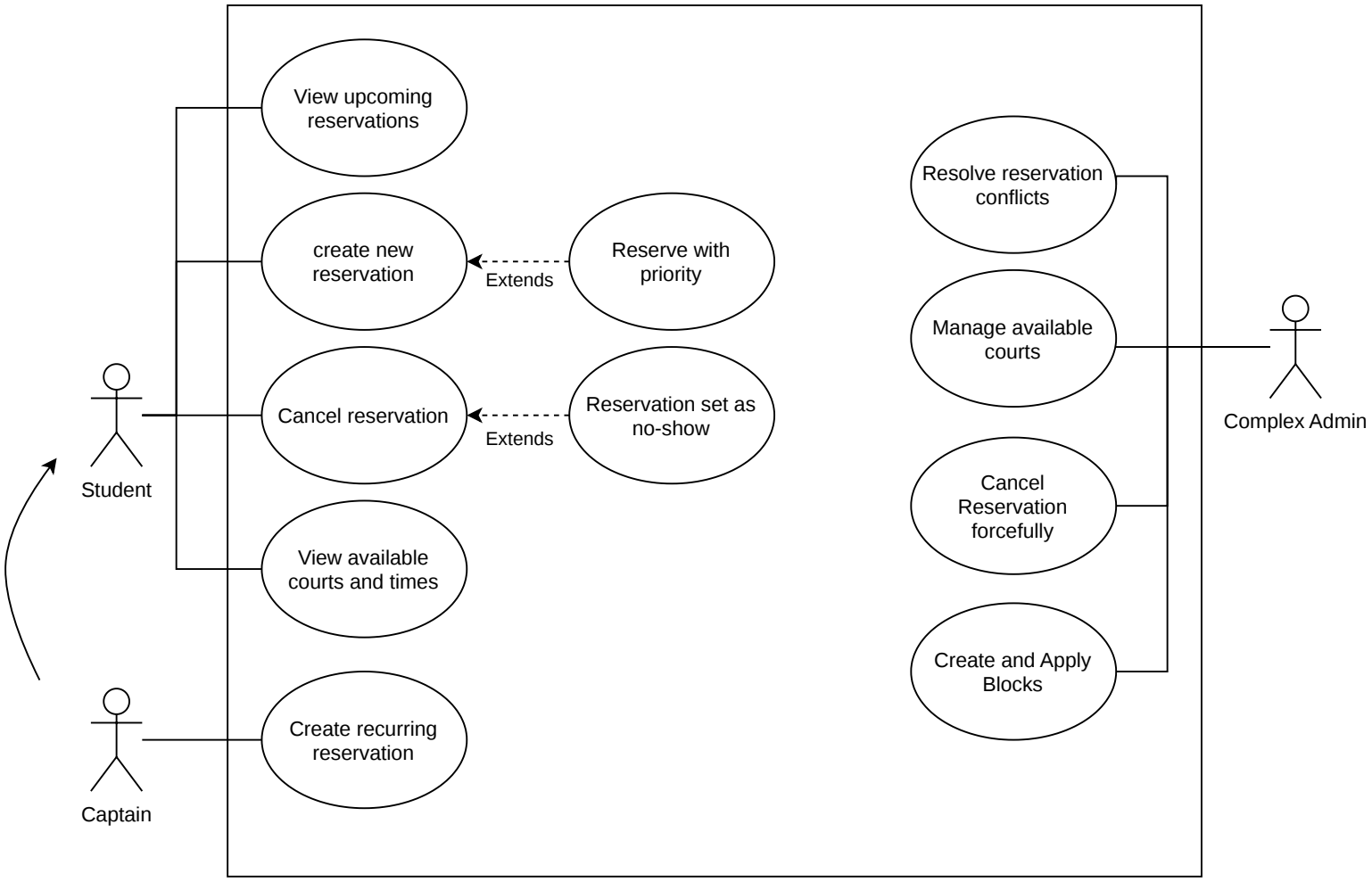
- RF-17: The system must manage reservations across different campuses, each of which may have different available times and notification systems.
- RF-18: The system must store configurable system values, including the priority-reservation window and cancellation window, in a single centralized configuration location.

- RNF-01: Under normal operating conditions, the system should respond to court-availability, reservation, and cancellation requests within two seconds.
- RNF-02: The system should be available at least 99% of the time during the university's normal operating hours.

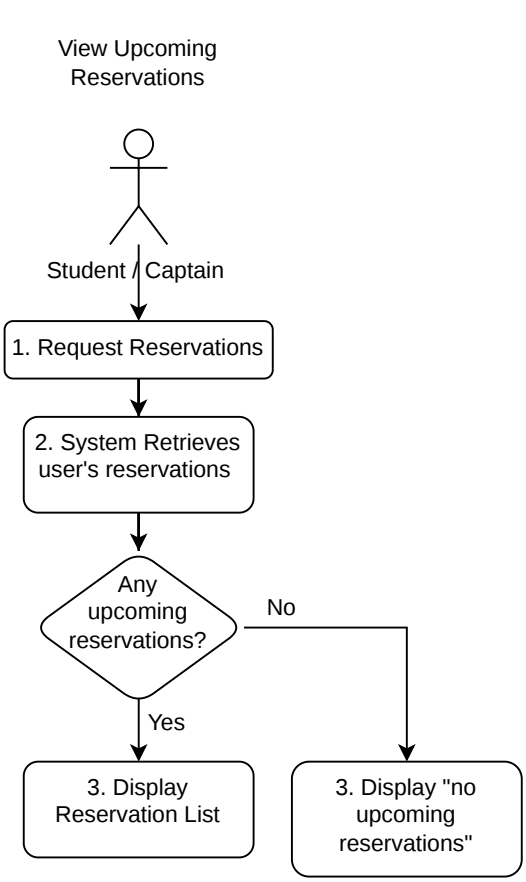
- RNF-03: The system must authenticate users using their university credentials.
- RNF-04: Students may view and cancel only their own reservations. Sports Complex administrators may manage courts and resolve reservation conflicts.

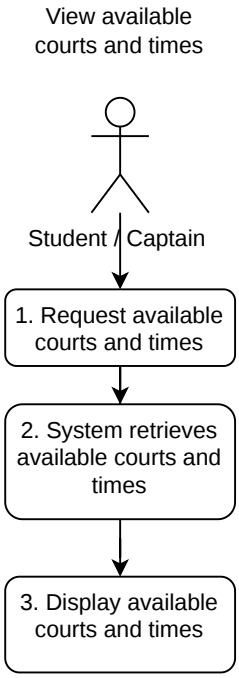
- RNF-05: The system must record reservation creation, cancellation, no-show, completion, and administrative conflict-resolution events, including the responsible user and timestamp.

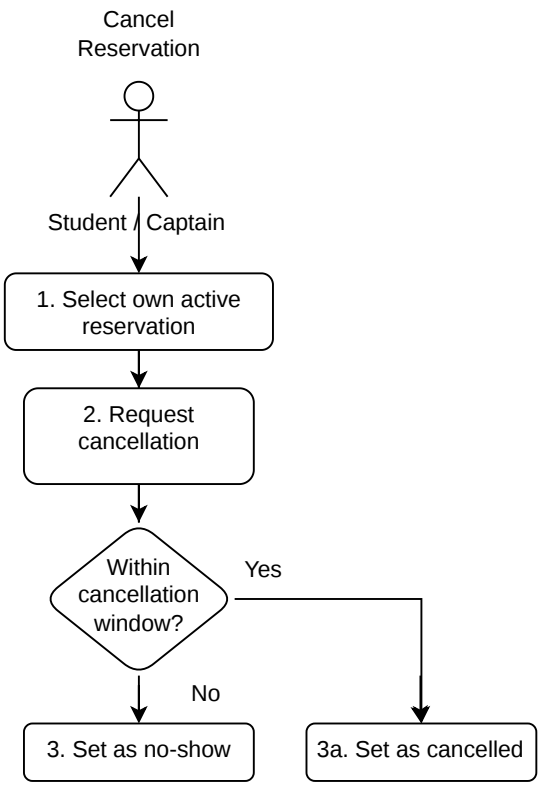
- RNF-06: The system must prevent two users from successfully reserving the same court for overlapping time periods, including when requests are submitted concurrently.
- RNF-07: When two overlapping reservations arrive simultaneously, the system must prioritize official team captains over students. If users have the same priority, it must favor the user with fewer no-shows. If they are still tied, it must favor the user with fewer recent reservations. If they remain tied, the system must use the server-side request sequence; if the requests are indistinguishable, it may use an auditable random selection.

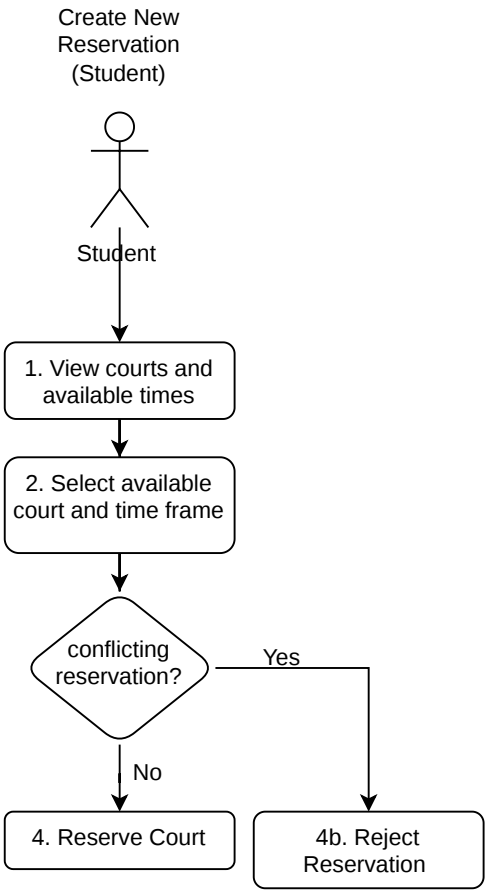


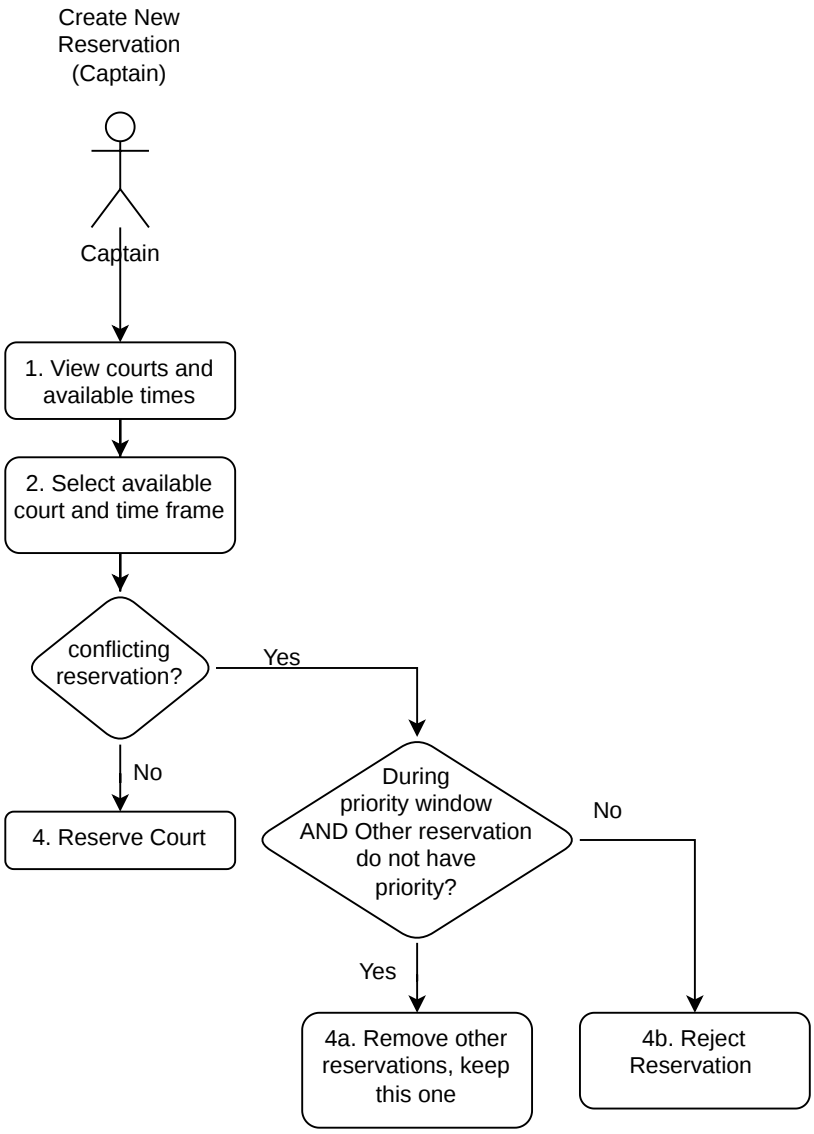
Preconditions	
View Future Reservations: - Logged in as student	Manage available courts - Logged in as admin
Request new reservation - Logged in as student	Resolve reservation conflicts - Logged in as admin
Cancel Reservation - Logged in as student	Cancel Reservation forcefully - Logged in as admin
Reserve with priority - Logged in as Captain	Create and Apply Blocks - Logged in as admin
Reservation set as no-show - If student: Cancel less than 2 hours before reservation start - If captain: Cancel less than 24 hours before reservation	
Create Recurring reservation - logged in as captain - No conflicts with other reservations	


View Upcoming Reservations
Student / Captain
1. Request Reservations
2. System Retrieves user's reservations
Any upcoming reservations?
No
Yes
3. Display Reservation List
3. Display "no upcoming reservations"

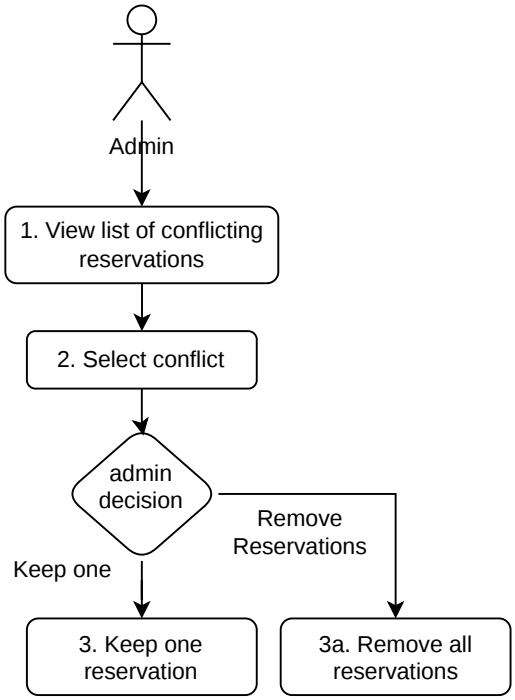

View available courts and times
Student / Captain
1. Request available courts and times
2. System retrieves available courts and times
3. Display available courts and times


Cancel Reservation
Student / Captain
1. Select own active reservation
2. Request cancellation
Within cancellation window?
Yes
No
3. Set as no-show
3a. Set as cancelled

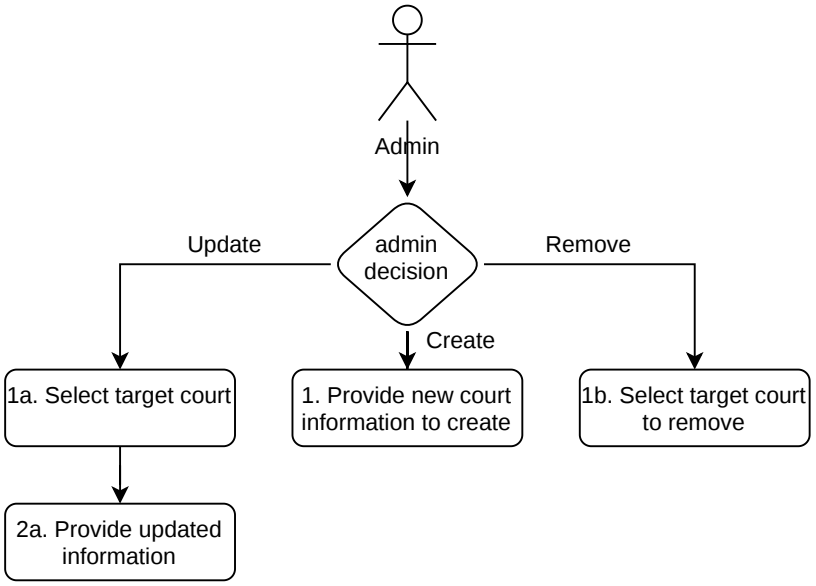

Create New Reservation (Student)
Student
1. View courts and available times
2. Select available court and time frame
conflicting reservation?
Yes
No
4. Reserve Court
4b. Reject Reservation


Create New Reservation (Captain)
Captain
1. View courts and available times
2. Select available court and time frame
conflicting reservation?
Yes
No
4. Reserve Court
During priority window AND Other reservation do not have priority?
No
Yes
4a. Remove other reservations, keep this one
4b. Reject Reservation

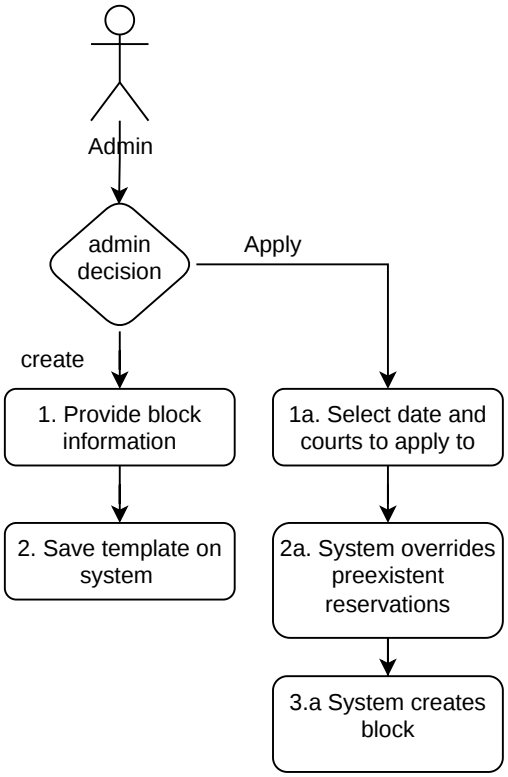
Resolve Reservation conflicts



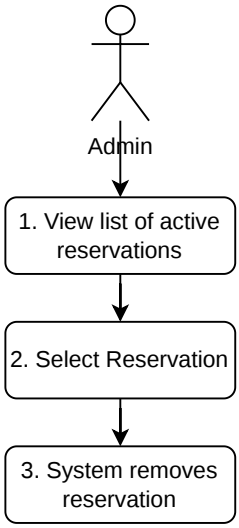
Manage Courts

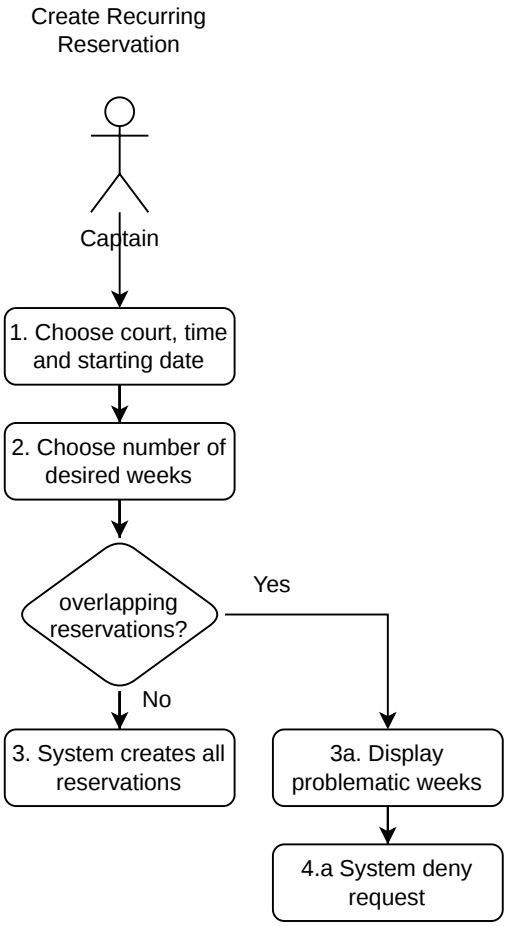


Manage Blocks

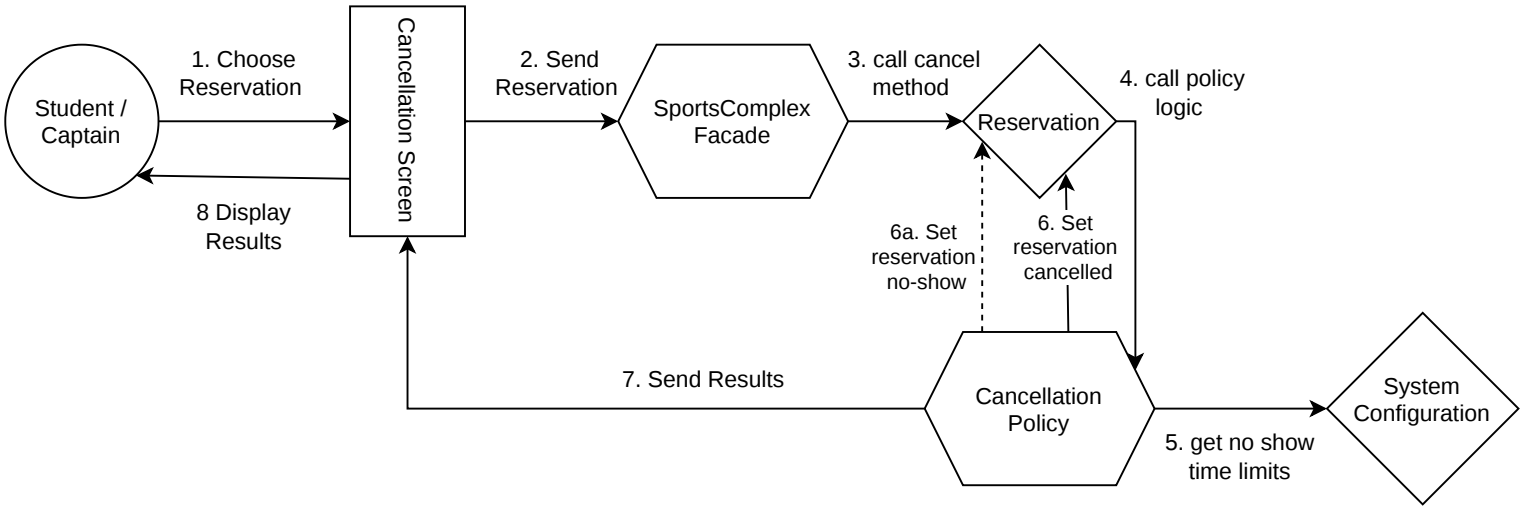


Remove reservation forcefully

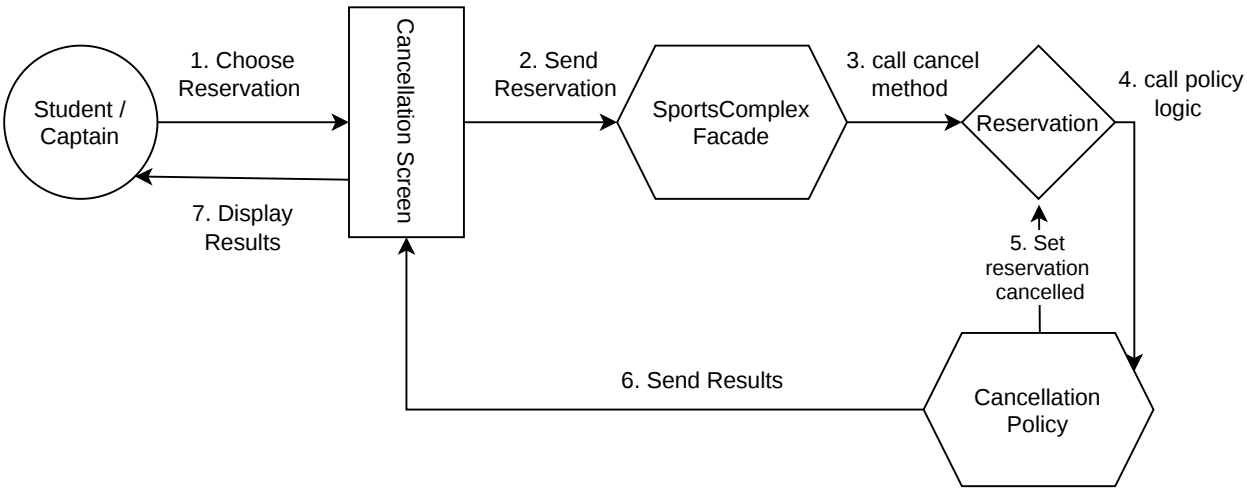



Create Recurring Reservation
Captain
1. Choose court, time and starting date
2. Choose number of desired weeks
overlapping reservations?
Yes
No
3. System creates all reservations
3a. Display problematic weeks
4.a System deny request

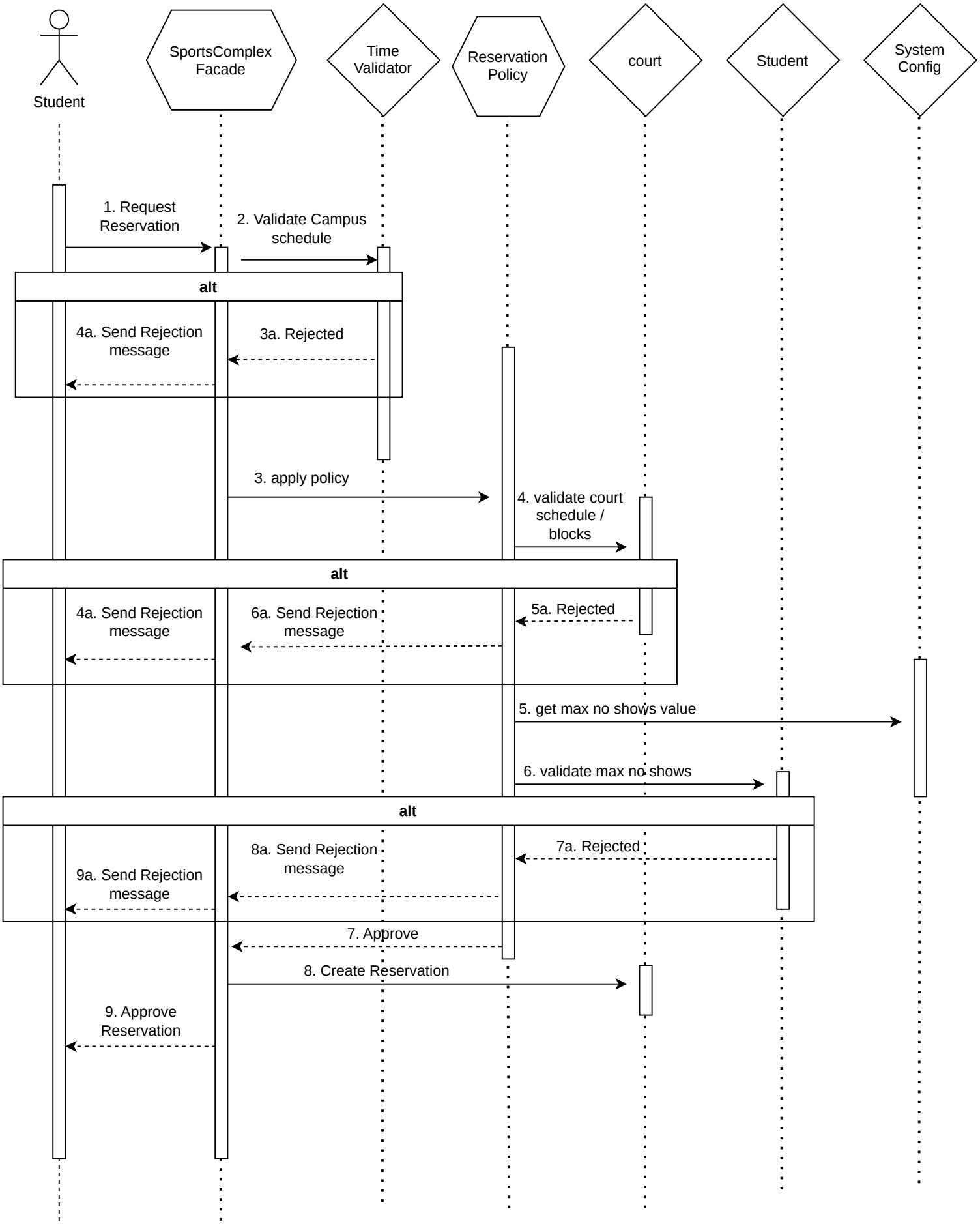
Cancel a Reservation (Student/Captain)



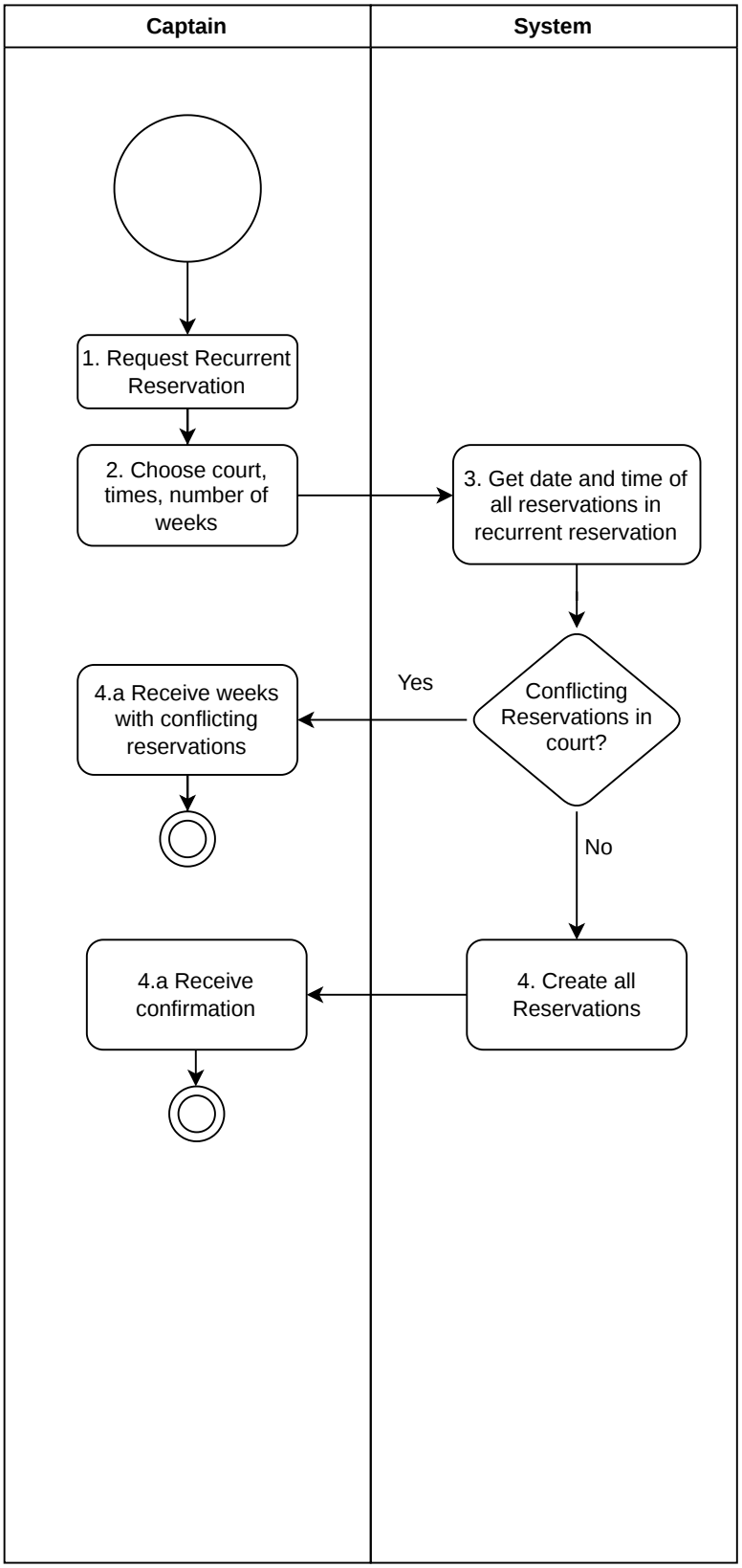
Cancel a Reservation Admin (Forced)



Create Reservation



Recurrent Reservation



Reservation State Machine

